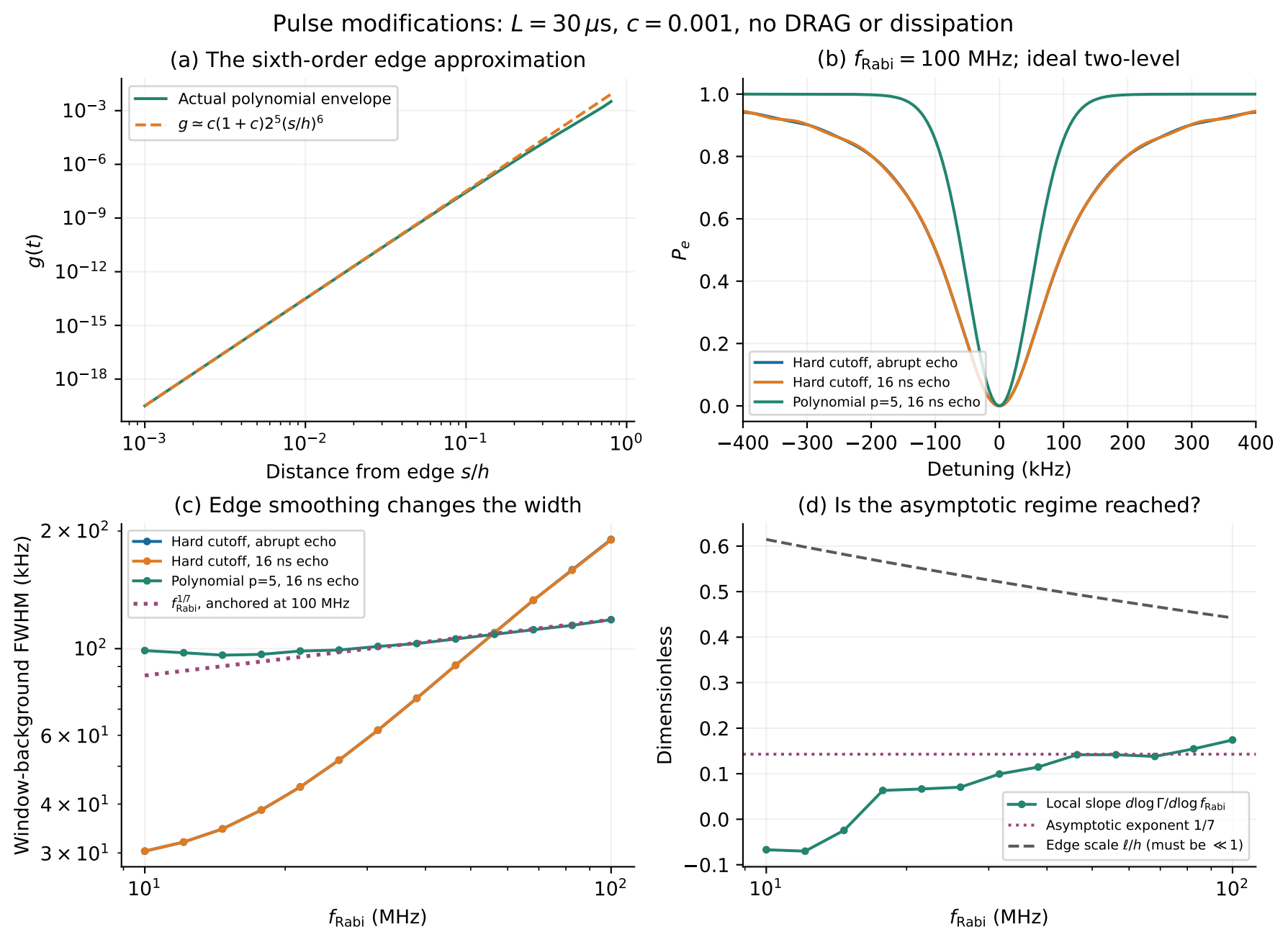
Pulse modifications: $L = 30\,\mu$s, $c = 0.001$, no DRAG or dissipation
(a) The sixth-order edge approximation
Actual polynomial envelope
$g \simeq c(1+c)2^5(s/h)^6$
$g(t)$
Distance from edge s/h
(b) $f_{\mathrm{Rabi}} = 100$ MHz; ideal two-level
P_e
Hard cutoff, abrupt echo
Hard cutoff, 16 ns echo
Polynomial p=5, 16 ns echo
Detuning (kHz)
(c) Edge smoothing changes the width
Hard cutoff, abrupt echo
Hard cutoff, 16 ns echo
Polynomial p=5, 16 ns echo
$f_{\mathrm{Rabi}}^{1/7}$, anchored at 100 MHz
Window-background FWHM (kHz)
f_{Rabi} (MHz)
(d) Is the asymptotic regime reached?
Dimensionless
Local slope $d\log\Gamma/d\log f_{\mathrm{Rabi}}$
Asymptotic exponent 1/7
Edge scale ℓ/h (must be $\ll 1$)
f_{Rabi} (MHz)